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Leveraging homogeneity-fusion-based TCN for enhanced batch residential load forecasting

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ABSTRACT

Electricity load forecasting is essential for maintaining the stability and efficiency of power systems. Factors such as geography, climate, and user behavior patterns often introduce inherent correlations among the energy consumption patterns of different residential users. However, most existing methods primarily emphasize the temporal characteristics of historical loads, while the modeling of homogeneity features across residential users remains insufficient, which constrains predictive accuracy. To address this challenge, this paper proposes a novel forecasting framework that employs a Multi-Head Attention mechanism to extract homogeneous features from multiple residential load and integrates them into the consumption data. The enriched data is subsequently processed by a Temporal Convolutional Network to extract temporal dependencies. The proposed approach leverages a Multi-Head Attention mechanism for batch load forecasting and enables the simultaneous prediction for all residential units in a single process while preserving their unique load patterns. Through more accurate load forecasting, the method presented in this paper provides robust support for building energy management, helping to optimize energy usage and enhance overall building energy efficiency. Experimental results show that the proposed model surpasses benchmark methods in both accuracy and generalization. In particular, the Mean Absolute Error is reduced by 15.5%, 9.74%, and 15.92% compared with LSTM, Informer, and CNN-GRU, respectively. These results confirm the effectiveness of the proposed method and highlight its potential for residential load forecasting.

1. Introduction

Power load forecasting plays an indispensable role in ensuring the safe, stable, and economical operation of modern power systems [1]. Accurate load prediction not only provides essential support for power dispatch but also constitutes a fundamental prerequisite for electricity market transactions and demand response [2]. Residential electricity consumption constitutes a substantial portion of overall power demand, and its fluctuations exert a direct influence on the supply-demand balance and operational efficiency of the power grid. Consequently, achieving high-precision residential load forecasting is essential for enhancing the flexibility and reliability of power system operations [3].

Residential load forecasting faces two major challenges in practical applications: batch processing and modeling homogeneity features. The first challenge stems from the exponential growth of residential load data [4,5]. Facing massive data volumes, regardless of whether the underlying system architecture is centralized or distributed, it is highly impractical to perform high-frequency, real-time data acquisition and model updates for each individual residential unit. Therefore, introducing a batch processing method that processes data within predetermined time windows can significantly enhance computational efficiency ,making it suitable for large-scale regional load forecasting. However, the significant variations in electricity consumption habits and climate responsiveness among different residences lead to markedly distinct load curve characteristics. This heterogeneity presents a major challenge, as a single unified model struggles to simultaneously optimize overall forecasting performance and accurately capture the unique characteristics of individual residences.

The second challenge is modeling homogeneity features. Factors such as geographical environment, climatic conditions, and user behavior patterns introduce inherent correlations among the energy consumption patterns of different residential users [6]. For example, homes in

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the same neighborhood may exhibit similar load consumption patterns due to similar temperatures and holiday effects. If the behavioral patterns of people in two homes located in different neighborhoods are very similar, those two homes' load consumption patterns may be more similar to each other than to other homes in the same neighborhood. In this study, these characteristics influencing residential energy consumption patterns are collectively defined as "homogeneity features". Consequently, the effective exploration and modeling of these homogeneity features are of great importance. However, most existing residential load forecasting methods have certain limitations in the following two aspects. First, some studies primarily focus on temporal modeling, while overlooking the exploration and modeling of homogeneous characteristics across residential units [7]. Second, other studies attempt to capture these homogeneous characteristics by explicitly incorporating external factors–such as climate, geographic location, and user behavior–into the input data. However, this approach generally models only partial homogeneity. And these external factors are often difficult to obtain and inherently subject to high levels of uncertainty [8].

To address these challenges, this paper proposes an MHA-TCN-W model for residential load forecasting. The proposed model can adaptively learning homogeneous features among different residences directly from historical load data through non-Euclidean pairwise correlations. In addition, the model enables batch processing while preserving the distinctive characteristics of individual residences. The main contributions of this study are summarized as follows:

(1) To effectively model the homogeneous features that influence energy consumption patterns, this study proposes a novel MHA-TCN-W load forecasting framework. The MHA mechanism extracts homogeneous features from multiple residential load and integrates them into the consumption data. Subsequently, the TCN extracts temporal features from the enriched data, thereby improving the accuracy of residential load forecasting.
(2) Considering the requirements for practicality and efficiency in large-scale applications, this study proposes a batch load forecasting method based on MHA. The proposed approach enables simultaneous prediction of all residential units in a single process while preserving their unique load patterns. It not only resolves the inefficiency associated with independent per-household forecasting, but also overcomes the limitations of traditional unified models in capturing individual characteristics.

The remainder of this paper is structured as follows. Section 2 reviews related work and motivates our research. Section 3 introduces the proposed MHA-TCN-W load forecasting algorithm. This is followed by simulation studies in Section 4. Finally, Section 5 concludes the paper.

2. Related work and motivation

This section first reviews load forecasting studies across various scenarios. It then examines research progress in residential load forecasting.

2.1. Load forecasting in different scenarios

In industrial park scenarios, load forecasting faces significant complexity. The power demand of industrial users often exhibits high-frequency and large-amplitude fluctuations driven by factors such as production schedules and equipment start-up or shutdown operations [9,10]. Moreover, significant differences exists among different industries, leading to considerable disparities in load curve characteristics across user categories [11].

For commercial building scenarios, load forecasting is characterized by high dependence on external factors and significant load fluctuations. Reference [12] optimizes input features through cross-correlation functions and proposes an improved LSTM model (ILSTM) to preserve key information. And it employs the DWDAdam optimizer for parameter tuning of the ILSTM.

In residential community scenarios, load forecasting faces substantial uncertainty and complexity. The stochastic and irregular nature of user behavior patterns [13] makes it difficult for forecasting models to accurately capture individual consumption characteristics. Moreover, residential loads are highly sensitive to external factors such as electricity pricing, weather conditions, and holidays [14,15].

In the three aforementioned scenarios, the deployment of load forecasting typically follows two types of system architectures: centralized [16] and distributed [17]. However, both architectures exhibit certain limitations when combined with traditional computational modes. Under a centralized architecture, while the traditional batch processing mode can enable unified management and improve overall prediction performance, it often ignores the variability among individual residences. Under a distributed architecture, while the common per-household real-time processing mode can preserve personalized features, it faces challenges such as low user cooperation and insufficient computational efficiency.

2.2. Review of residential load forecasting

In recent years, research on residential load forecasting has achieved remarkable progress. Existing approaches can be broadly divided into two main categories: (1) deep learning methods based on neural network architectures, and (2) hybrid modeling approaches. In recent years, recurrent neural networks [18], long short-term memory networks (LSTM) [19], and gated recurrent units [20] have been extensively applied to residential load forecasting tasks. Temporal convolutional network (TCN), an advanced variant of convolutional neural networks(CNN), have emerged as one of the most promising deep learning frameworks for time-series forecasting [21,22]. Reference [23] applied TCN to residential load forecasting tasks, effectively addressing issues such as low efficiency in modeling long-term sequence dependencies and unstable training in traditional recurrent neural networks. Compared with single-model architectures, hybrid models are capable of integrating the advantages of multiple approaches to achieve superior predictive performance [24,25]. The incorporation of attention mechanisms has emerged as a crucial technical pathway [26–28].

To further improve the accuracy of load forecasting, an increasing number of studies have begun to consider external factors that influence user energy consumption patterns, such as climatic factors [29,30], geographical factors [31], and user behavioral patterns [32]. These studies have demonstrated that explicitly incorporating external features into forecasting models can effectively enhance predictive performance. Reference [33] improves the accuracy of residential load forecasting by inputting the energy consumption sequences of major household appliances into an LSTM framework, thereby capturing user behavioral pattern features more effectively.

However, the aforementioned prediction models generally face two limitations. First, most existing studies overlook the potential inherent correlations among different residences. Second, they all depend on the explicit input of external features.

To overcome the bottlenecks of external features acquisition and the limitations of isolated per-household modeling, an increasing number of recent studies have shifted towards cross-series dependency modeling [34]and multi-household joint modeling [35]. Reference [36] proposed a global-local forecasting framework employing a hybrid attention Transformer and a Gaussian process. This framework utilizes task attention and temporal attention to capture the cross-sequence coupling among multi-energy loads and the single-sequence temporal dependencies, respectively.

Having established the advantages of cross-sequence dependency modeling, researchers have further introduced Graph Neural Networks (GNN) [37]. Reference [38] constructs temporal and spatial pyramidal graphs to extract multi-household dynamic spatial correlations along with long- and short-term temporal dependencies. It effectively enhances multi-step forecasting accuracy through gated fusion and a trans-

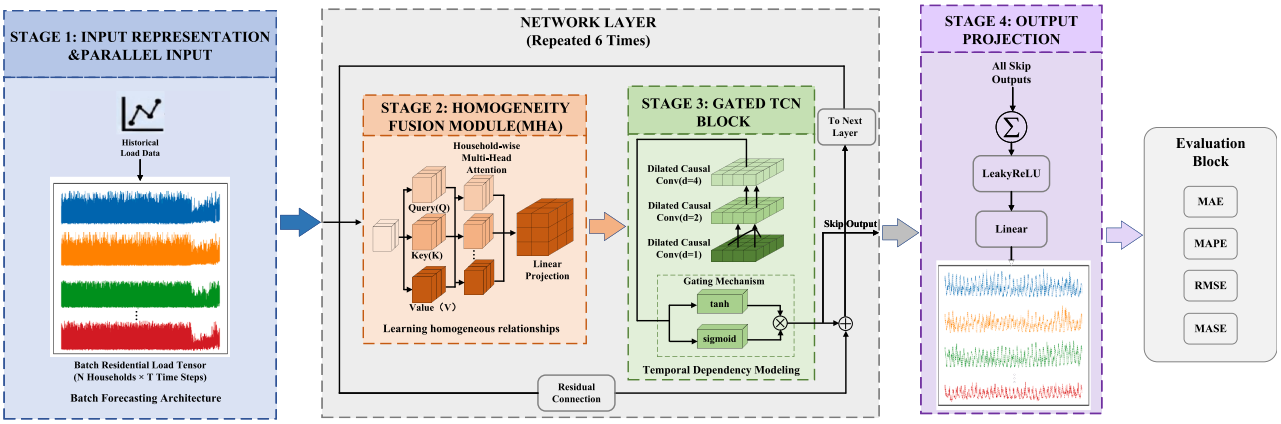


Fig. 1. MHA-TCN-W based load prediction framework.

form attention block. Addressing the missing data scenarios in practical applications, reference [39] extracts cross-sequence spatial dependencies across multiple nodes via a graph architecture. Coupled with adaptive multi-scale temporal modeling, it significantly improves the accuracy and robustness of load forecasting under incomplete data conditions.

3. Method

This section introduces the MHA-TCN-W framework. The overall structure of the proposed model is illustrated in Fig. 1. It consists of the following four stages.

Stage 1: Historical load sequences from multiple residences are subjected to dimensionality transformation to obtain a unified feature representation.

Stage 2: The feature representations obtained in Stage 1 are fed in parallel into the MHA module. Within each feature space, each residence is treated as the target building and computes dot-product similarity scores with the remaining candidate buildings. Subsequently, the similarity scores are normalized and converted into attention weights, which are used for weighted fusion to generate a residence feature representation that integrates homogeneous features. Compared with the original features, the enhanced representation contains richer information and exhibits improved discriminative power and generalization capability. Moreover, because the similarity calculation is anchored to the target building's own features, its unique characteristics are fully preserved during the fusion process.

Stage 3: The enhanced feature representation are fed into the Gated TCN module. The TCN employs dilated convolutions to extract multi-scale temporal dependencies. In addition, a gating mechanism is introduced to adaptively regulate information flow, thereby enhancing the model's capability to capture peak-off-peak load variations.

Stage 4: The outputs from each layer are propagated and aggregated through a skip-connection mechanism, followed by Leaky-Relu activation and a linear mapping to produce the final prediction results.

3.1. Multi-head attention

To effectively capture the inherent relationships in energy consumption patterns among different residences, this paper introduces the MHA mechanism within the proposed framework. MHA was originally introduced by Vaswani et al. [40] in the Transformer architecture. In the context of residential load forecasting, the energy consumption behaviors of different residences are often influenced by similar external factors, resulting in a certain degree of homogeneity. The MHA mechanism models and integrates these homogeneous features by computing attention weights among residences in the feature space, thereby adaptively identifying and quantifying both their similarities and differences. The structure of the MHA module is depicted in Fig. 2. Its core principle involves computing a correlation weight matrix through the interaction of Query (Q), Key (K), and Value (V), followed by a weighted aggregation of input features based on the learned attention weights.

It should be emphasized that the proposed model aims to extract and integrate homogeneity features among different residences from historical load data; therefore, the model's input consists solely of the raw load sequences for all residences. Suppose there are N buildings. To comprehensively capture the fluctuation patterns of the time series, the model flattens and concatenates the consecutive time-series features from C channels within a sliding window of length T, resulting in a composite feature vector with an embedding dimension F ($F = C \times T$). Let $X \in R^{N\times F}$ denote the overall trajectory of each building within the current time window. MHA projects input features into a high-dimensional attention space via a standard linear projection layer. For the hth attention head, the Query, Key, and Value representations are defined respectively as:

$$Q_i^{(h)} = X_i W_Q^{(h)}, \quad K_j^{(h)} = X_j W_K^{(h)}, \quad V_j^{(h)} = X_j W_V^{(h)} \tag{1}$$

where $W_Q^{(h)}, W_K^{(h)}, W_V^{(h)} \in R^{F\times d}$ denotes the linear mapping parameter, $d = F/H$ represents the dimensionality of a single attention head, and h indicates the total number of attention heads.

Under the h-th attention head, the similarity between the target building i and each candidate building j is computed using a scaled dot-product mechanism and subsequently normalized to obtain the corresponding attention weights. The computation is formulated as follows:

$$A_{ij}^{(h)} = \mathrm{softmax}\left(\frac{\frac{Q_i^{(h)}\left(K_j^{(h)}\right)^{\top}}{\sqrt{d}} + \Delta_{ij}}{\tau}\right), \quad i,j \in \{1,\ldots,N\} \tag{2}$$

where, Δ_{ij} represents the (i,j)th element of the learnable bias matrix, using to adaptively adjust the relationships between buildings. τ is a learnable temperature parameter, which is designed to control the smoothness of the attention distribution. A smaller τ leads to a sharper attention weight distribution and a larger τ results in a smoother attention distribution. softmax$(\cdot)$ is applied along the candidate building j dimension, ensuring that the attention weights for each target building i sum to 1.

The output of the h-th attention head for the target building i is then obtained by performing a weighted aggregation of the features of all candidate buildings.

$$H_i^{(h)} = \sum_{j=1}^{N} A_{ij}^{(h)} V_j^{(h)} \tag{3}$$

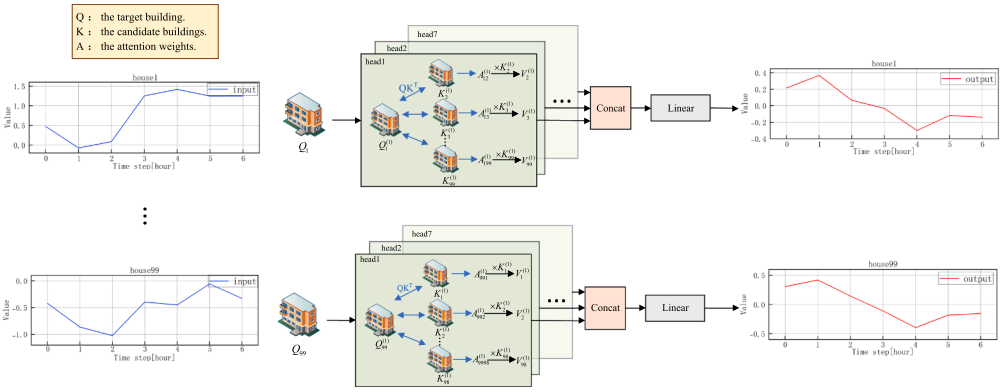


Fig. 2. Block diagram of the MHA architecture.

The final representation of target building i is obtained by concatenating the aggregated vectors from all attention heads:

$$Y_i = \text{concat}(H_i^{(1)}, H_i^{(2)}, \ldots, H_i^{(H)})W_O \tag{4}$$

where $W_O \in \mathbb{R}^{F\times F}$ denotes the output projection matrix.

To address the issues of representation homogenization and over-smoothing that are prone to occur during inter-building feature aggregation, this paper introduces regularization and control mechanisms within the MHA module. Specifically, a learnable temperature parameter τ is introduced during the attention weight calculation process to adjust the smoothness of the attention distribution. Furthermore, an Attention Dropout strategy is employed after the generation of the attention weight matrix. By randomly suppressing the associative connections between portion of buildings during the training process, it reduces the model's over-reliance on specific connection patterns.

3.2. Gated temporal convolution network

In this study, a TCN-based model is employed to extract temporal dependency features across nodes. In addition, a gating mechanism [41] is integrated into the TCN to enhance its ability to capture complex nonlinear relationships.

1) Dilated causal convolution

By inserting gaps between kernel elements, dilated convolutions achieve exponential growth of the receptive field, thereby avoiding the computational burden of excessively deep network structures. The calculation formula is expressed as follows:

$$y(t) = \sum_{i=0}^{k-1} w(i) \cdot X(t - d \cdot i) \tag{5}$$

where X denotes the input, $y(t)$ represents the output at time step t, $w(i)$ indicates the convolution kernel parameters, k specifies the kernel size, and d corresponds to the dilation factor. By stacking multiple dilated convolution layers, the model can capture long-range temporal dependencies within a finite network depth.

Fig. 3 illustrates an example of a dilated causal convolution, where the input historical load sequence has a length of 7, the convolution kernel size is 2, and the dilation factor sequence is $(1, 2, 4)$. It can be observed that as both the number of network layers and the dilation factors increase, the model's receptive field expands accordingly, allowing it to capture historical dependencies over longer time horizons.

2) Gated mechanism

A gating mechanism is incorporated to enhance the model's nonlinear modeling capability for time-dependent features. This mechanism generates candidate feature representations and gating weights through two parallel convolutional pathways, which are then combined via element-wise multiplication. Let X denote the input to the TCN; the

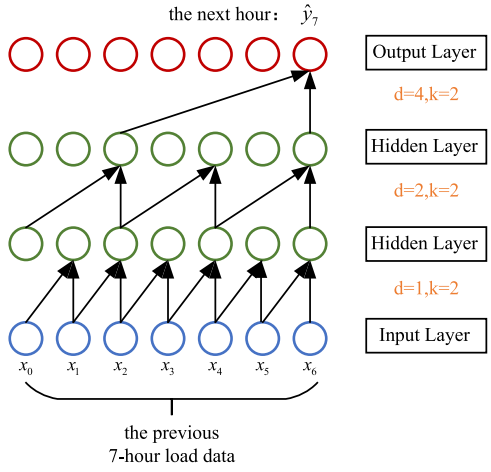


Fig. 3. Illustration of dilated causal convolution.

corresponding gated TCN output O can be expressed as:

$$O = \tanh\left(\text{Conv}_f(X)\right) \odot \sigma\left(\text{Conv}_g(X)\right) \tag{6}$$

where $\text{Conv}_f(\cdot)$ and $\text{Conv}_g(\cdot)$ denote two distinct convolutional transformations, $\tanh(\cdot)$ produces the candidate features, $\sigma(\cdot)$ generates the gating weights, and $\odot$ denotes element-wise multiplication. This structure adaptively regulates the flow of information, thereby facilitating the capture of complex nonlinear dynamics within load sequences.

3) WaveNet

In this study, the WaveNet framework is employed to stack multiple network layers, with a skip-connection mechanism used to transmit outputs of each layer to external modules. The outputs from multiple layers are then integrated through Leaky Relu activation functions and linear layers to produce the final prediction results.

WaveNet was originally proposed by DeepMind for the generation of raw audio waveforms [42]. Its core principle lies in employing dilated convolutions within a TCN framework to construct deep architectures capable of capturing long-term dependencies in a non-recursive manner. WaveNet models can be extended to any other type of time series prediction problem. In wind power forecasting [43], WaveNet's expansive causal convolution properties enable a larger receptive field with fewer network layers, effectively extracting both long-term and short-term patterns from wind power time series data.

3.3. Forward pass of the model

To clarify the data flow and tensor transformation process within the proposed MHA-TCN-W architecture, its forward-pass mechanism is de-

tailed as follows. Given an initial historical load tensor $X \in R^{B\times C\times T\times N}$ (where B is the batch size, C is the number of channels, T is the number of time steps, and N is the number of buildings), the tensor is first subjected to permutation and reshaping operations to be transformed into a feature matrix $R^{B\times N\times F}$ (where $F = C \times T$). It is then input into the MHA module to extract and integrate homogeneity features between buildings.

Subsequently, to satisfy the "channels-first" input convention required by 1D convolutions, the model performs a format conversion on the MHA output. The attention output is reshaped into $R^{B\times N\times T\times C}$, permuted into $R^{B\times C'\times T\times N}$, and finally flattened by merging the batch and node dimensions to construct a TCN input tensor of shape $R^{(B\cdot N)\times C'\times T}$.

This tensor is then fed into the dilated causal convolution module equipped with a gating mechanism. Two parallel dilated causal convolution branches independently generate candidate features and gating weights, realizing gating interaction through point-wise multiplication. After completing the temporal convolution, the hidden features are restored to the four-dimensional format $R^{B\times C_{\text{hidden}}\times T\times N}$ for residual connections. The model employs a skip connection mechanism to extract and retain the output features from each network layer.

After traversing all network layers, the features collected from each layer are directly aggregated using element-wise summation. Finally, the aggregated features are mapped sequentially through LeakyRelu activation function and 2D convolution layer, directly generating the final load forecasting result $\hat{Y} \in R^{B\times 1\times N}$.

4. Simulation

In this section, the MHA-TCN-W model was implemented using the PyTorch framework in a Python environment. Model training and testing were conducted on a laptop equipped with an NVIDIA GeForce RTX 3050 GPU, an AMD Ryzen 7 6800HS CPU, and 16 GB of memory.

4.1. Data source and evaluation index

This section introduces the two datasets used in this study. Dataset 1 comprises historical hourly electrical load data for 99 residential buildings in London [44], spanning from December 2012 to December 2013. Dataset 2 consists of hourly electrical load data for 50 residential buildings in Australia from June 2013 to December 2013 [45]. Compared with Dataset 1, Dataset 2 is approximately half in size in terms of both the number of residential buildings and the temporal coverage. Both datasets were chronologically divided into training, validation, and test sets in a ratio of 7:2:1.

To reduce the impact of scale disparities on model training, the training, validation, and test sets were normalized on a house-by-house basis using the means and standard deviations of the corresponding nodes calculated from the training set. The normalization is formulated as follows:

$$\hat{x}_{ij} = \frac{x_{ij} - \text{mean}(X_{i-\text{train}})}{\text{std}(X_{i-\text{train}})} \tag{7}$$

where x_{ij} denotes the original load consumption of building i at time step j, $\hat{x}_{ij}$ represents the normalized value, and $\text{mean}(X_{i-\text{train}})$ and $\text{std}(X_{i-\text{train}})$ correspond to the mean and standard deviation of building i's training set, respectively. In addition, this study employs the historical load data from the preceding 7 h to forecast the electricity consumption of the next hour.

During model training, high-load samples are often prone to underfitting. To mitigate this issue, this study incorporates a weighted MAE loss function into the training objective, augmented with variance-matching terms.

Samples with load values exceeding a predefined threshold are assigned higher loss weights, guiding the model to place greater emphasis on accurate prediction during high-load periods throughout the training process. The mathematical formulation is expressed as follows:

$$L_{\text{wMAE}} = \frac{1}{BNT}\sum_{b=1}^{B}\sum_{i=1}^{N}\sum_{t=1}^{T}\omega_{b,i,t}\left|\hat{y}_{b,i,t} - y_{b,i,t}\right| \tag{8}$$

where B denotes the number of samples within a batch,which is set to 64; N represents the number of nodes, comprising 99 for Dataset 1 and 50 for Dataset 2; and T indicates the number of time steps per sample, which is configured to 7. Furthermore, $\hat{y}_{b,i,t}$ and $y_{b,i,t}$ correspond to the predicted and actual values, respectively, for the b-th sample, the i-th node, and the t-th time step.

The weight $\omega_{b,i,t}$ at each time point is defined as follows:

$$\omega_{b,i,t} = \begin{cases} \omega_{\text{high}} = \min\left(\dfrac{n_{\text{low}}}{n_{\text{high}} + \varepsilon}\cdot\lambda, \omega_{\max}\right), & \text{if } y_{b,i,t} > \tau \\ \omega_{\text{low}} = 1, & \text{otherwise} \end{cases} \tag{9}$$

where τ denotes the high-load threshold, defined as the 95th percentile value of the dataset. Specifically, it is set to 8.6 for Dataset 1 and 9.9 for Dataset 2. λ represents the weight scaling factor and is assigned a value of 0.4. $\omega_{\max}$ specifies the upper bound for high-load weights, thereby preventing model training crashes or prediction failures caused by abnormal weight values, and is set to 2. ε is introduced to avoid division by zero and is set to e^{-8}. When $\omega_{\text{high}} = 0$, it defaults to $\omega_{\text{high}} = 1$.

To prevent overly smoothed outputs or insufficient fluctuation in the model predictions, this study further incorporates a variance-matching term, defined as follows:

$$V_{\text{ar}_t}(\mathbf{y}_b) = \frac{1}{T}\sum_{t=1}^{T}\left(y_{b,t} - \bar{y}_b\right)^2, \tag{10}$$

$$L_{\text{var}} = \frac{1}{B}\sum_{b=1}^{B}\left(V_{\text{ar}_t}(\hat{\mathbf{y}}_b) - V_{\text{ar}_t}(\mathbf{y}_b)\right)^2 \tag{11}$$

where $V_{\text{ar}_i}(\mathbf{y}_b)$ denotes the true sequence variance of the b-th sample, which measures the overall fluctuation amplitude of that sequence. $y_{b,t}$ represents the true value of the b-th sample at time step t. $\bar{y}_b$ indicates the mean value of the b-th sample sequence. $V_{\text{ar}_i}(\hat{\mathbf{y}}_b)$ signifies the predicted sequence variance of the b-th sample.

In summary, the composite loss function employed during model training is defined as follows:

$$L = L_{\text{wMAE}} + \beta L_{\text{var}} \tag{12}$$

where β represents the balancing coefficient and is set to 0.2.

To comprehensively evaluate the predictive performance of different models, four evaluation index are employed in this study: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE) and Mean Absolute Scaled Error (MASE). The equations are as follows:

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t| \tag{13}$$

$$MAPE = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{14}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{15}$$

$$MASE = \frac{\frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|}{\frac{1}{n-1}\sum_{t=2}^{n}|y_t - y_{t-1}|} \tag{16}$$

where y_t denotes the actual value at time step t, $\hat{y}_t$ represents the corresponding predicted value at that time, and n indicates the total number of time steps.

4.2. Simulation scenarios

The effectiveness of the proposed MHA-TCN-W algorithm was evaluated under three scenarios. The hyperparameter settings of the MHA-TCN-W model are summarized in Table 1.

Scenario 1: Sensitivity analysis of model hyperparameters.

To evaluate the impact of the dilation factor sequence on the model's predictive performance, we conducted sensitivity analysis experiments. We configured two representative dilation factor sequences, $(1, 2, 1, 2, 1, 2)$ and $(1, 2, 4, 8, 16, 32)$, to compare and quantify its feature extraction capability and prediction accuracy under different receptive field scales.

Scenario 2: Validation of the effectiveness of each module in MHA-TCN-W.

To evaluate the performance of the MHA, the attention weight matrix was visualized and analyzed. Then, ablation experiments were conducted by removing the MHA module and the WaveNet module respectively, to quantitatively assess their contributions to the overall performance improvement of the model.

Scenario 3: Validation of the effectiveness of the MHA-TCN-W algorithm.

To evaluate the performance of MHA-TCN-W, four baseline models—LSTM, Informer, CNN-GRU, and Ada-GWN—are selected for comparison. In terms of input settings, all baseline models strictly adopt the same data split ratio, normalization strategy, and input and prediction window lengths as the proposed model. Regarding model architecture, each baseline follows its classical configuration: LSTM adopts the architecture described in [46], consisting of stacked LSTM layers and fully connected layers; Informer follows the standard settings outlined in its original work [47]; CNN-GRU employs three stacked 2D convolutional layers (with a kernel size of (2,1)) to extract local features, followed by GRU layers and fully connected layers to capture temporal dependencies; Ada-GWN adopts the recommended configuration in [48]. For hyperparameter tuning, the optimization of key parameters (e.g., hidden layer dimensions, number of layers, and convolution kernel sizes) follows the theoretical framework described in [49,50], using a grid search strategy within predefined ranges. All models are trained using the Adam optimizer, and an early stopping strategy based on the validation MAE is employed to prevent overfitting.

In this paper, four forecasting algorithms are chosen as the main baseline algorithms. Among them, LSTM is a classical RNN-based load forecasting model that is suited for capturing temporal dependencies in sequential data. Informer is capable of modeling global temporal dependencies through its sparse self-attention mechanism. However, both models primarily focus on temporal feature extraction and lack explicit mechanisms to capture homogeneous features among residential units. CNN-GRU introduces local convolutions to extract neighboring patterns, implicitly incorporating limited inter-residential information; however, its effectiveness depends on predefined node arrangements and simple association structures, which restricts its ability to represent complex homogeneous relationships. In contrast, Ada-GWN based on GNN, can capture inherent homogeneous feature dependencies among residential units. Therefore, comparisons with these baseline models are conducted to validate the effectiveness of the proposed MHA-TCN-W in residential load forecasting.

4.3. Simulation results for scenario 1

Table 2 presents the prediction results for Dataset 1 and Dataset 2 under different dilation factor sequences. Overall, the model achieves the best predictive performance on both datasets when the dilation factor sequence is set to (1, 2, 4, 1, 2, 4). Compared with the configuration (1, 2, 1, 2, 1, 2) featuring a smaller receptive field, it more effectively captures medium- to long-term temporal dependencies. In contrast to the exponentially increasing configuration (1, 2, 4, 8, 16, 32), it avoids

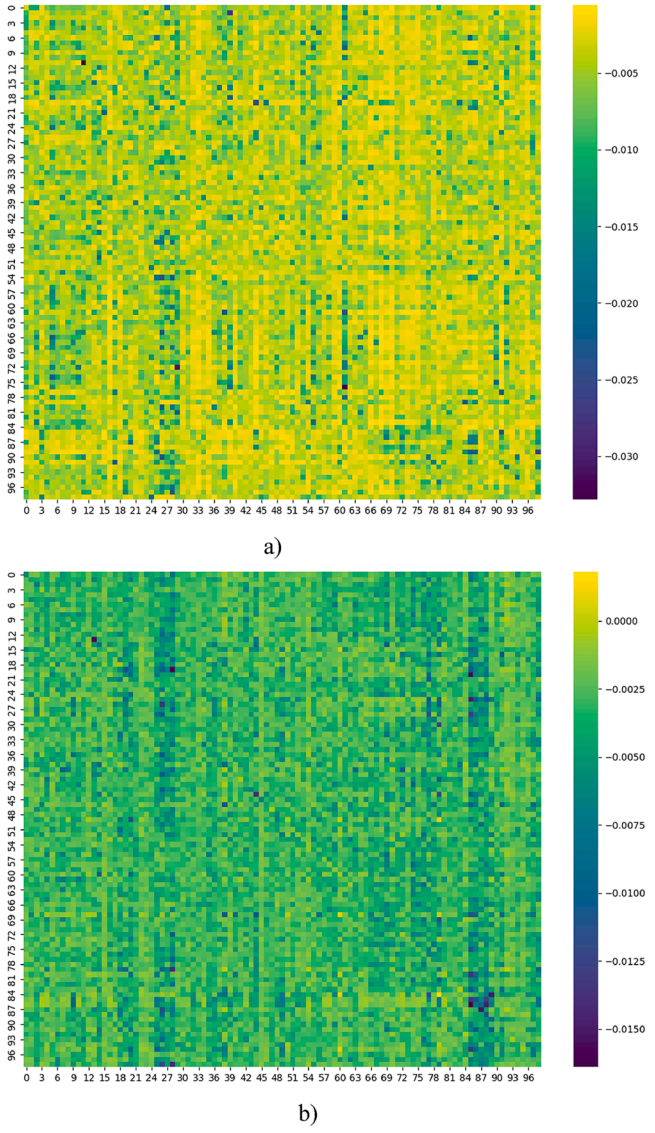


Fig. 4. Weight heatmap of dataset 1. a) Weight heatmap of the first layer for the first sample. b) Weight heatmap of the last layer for the first sample.

Table 1
Main parameters used in MHA-TCN-W.

Name	Value
Number of attention heads	7
MHA dropout rate	0.3
Sequence of dilation factor	(1,2,4,1,2,4)
Layer of WaveNet	6
Hidden dimension	32
Model dropout rate	0.1
Learning rate	0.0005
Batch size	64
Epoch number	100

Table 2
Calculation results under different sequences of dilation factors.

Dataset	Sequence of dilation factor	Metrics			
		MAE	MAPE(%)	RMSE	MASE
1	(1,2,4,1,2,4)	0.7718	18.90%	1.1416	0.3304
	(1,2,1,2,1,2)	0.7739	19.02%	1.1398	0.3313
	(1,2,4,8,16,32)	0.7742	19.08%	1.1391	0.3314
2	(1,2,4,1,2,4)	0.8360	24.96%	1.2578	0.4049
	(1,2,1,2,1,2)	0.8371	25.03%	1.2631	0.4054
	(1,2,4,8,16,32)	0.8396	25.68%	1.2648	0.4066

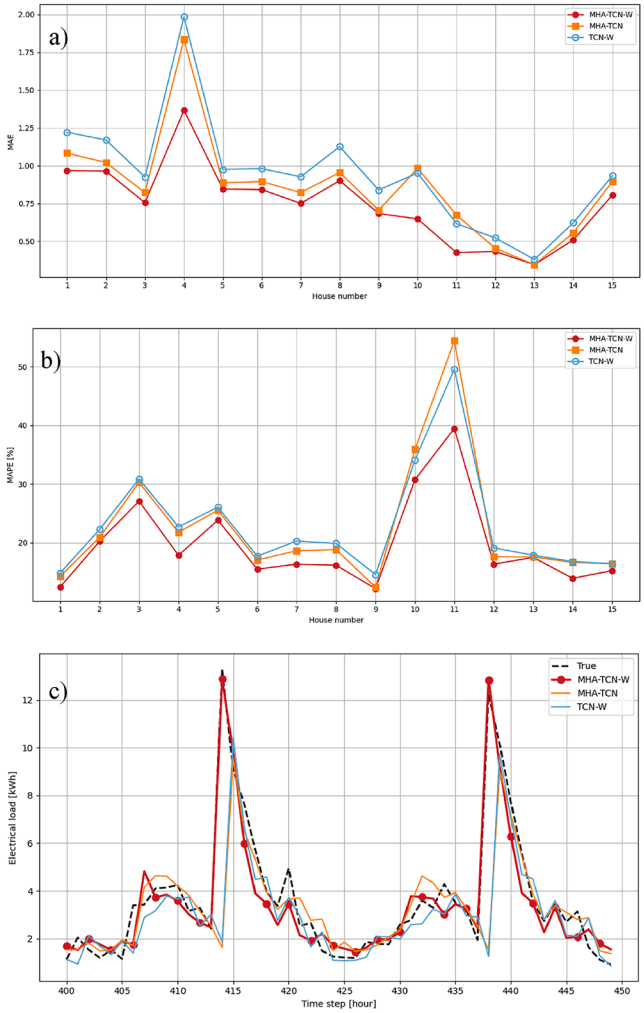


Fig. 5. Performance of models on dataset 1. a) MAE of 15 randomly selected buildings. b) MAPE of 15 randomly selected buildings. c) Visualization of the prediction results for the first building.

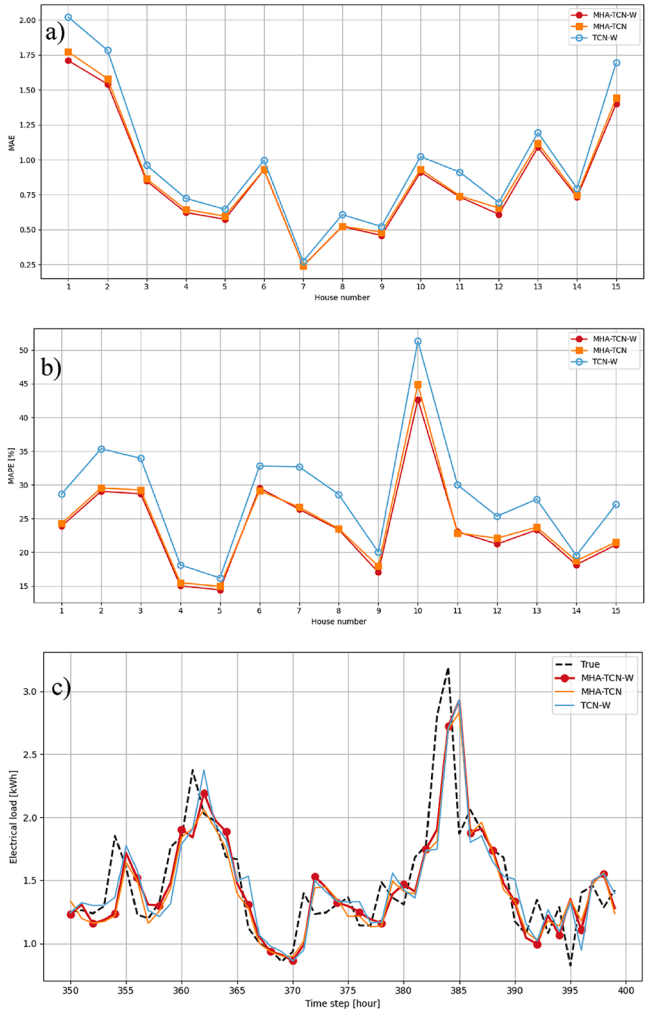


Fig. 6. Performance of models on dataset 2. a) MAE of 15 randomly selected buildings. b) MAPE of 15 randomly selected buildings. c) Visualization of the prediction results for the first building.

distant noise interference caused by an excessively large receptive field, and the dilution of critical short-term features.

For Dataset 1, when the dilation factor sequence is set to (1, 2, 4, 8, 16, 32), the MAE and MASE increase by 0.31% and 0.30%, respectively. Under the same configuration, the MAE and MASE for Dataset 2 increase by 0.43% and 0.42%, respectively. The performance fluctuations of both datasets across different configurations remain within 5%. This indicates that the proposed MHA-TCN-W model maintains good stability and robustness under variations in the dilation factor sequence.

4.4. Simulation results for scenario 2

For brevity, Fig. 4 illustrates the visualization of the attention weight matrix using Dataset 1 as a representative example. Fig. 4(a) shows the heatmap of the weight distribution for the first sample in the first layer of the attention mechanism. The overall weight distribution appears relatively dispersed, with no distinct differences in dependency relationships among residential units. In contrast, Fig. 4(b) displays the attention weight distribution for the same sample in the final layer. By this stage, the overall distribution exhibits significant changes: weights between certain residential units have markedly strengthened, revealing stronger homogeneous associations; conversely, weights between weakly correlated units have further diminished. This evolution pattern indicates that as the network depth increases, the MHA progressively filters and reinforces strong correlations among key nodes while attenuating irrelevant or weak connections. This process facilitates the adaptive integration of critical homogeneous features at higher representation levels.

Table 3
Calculation results of ablation study.

Dataset	Method	Metrics			
		MAE	MAPE(%)	RMSE	MASE
1	MHA-TCN-W	0.7718	18.90%	1.1416	0.3304
	MHA-TCN	0.8195	20.24%	1.2513	0.3508
	TCN-W	0.9130	22.39%	1.3861	0.3885
2	MHA-TCN-W	0.8360	24.96%	1.2578	0.4049
	MHA-TCN	0.8482	25.29%	1.2744	0.4108
	TCN-W	0.9560	29.93%	1.4190	0.4630

The results of ablation experiments are presented in Table 3. The findings indicate that the inclusion of the MHA and WaveNet modules improves model performance. Specifically, for Dataset 1, the MAE and MASE of MHA-TCN-W decrease by approximately 5.82% respectively, compared with MHA-TCN, and by 15.47%and 14.95%, respectively, compared with TCN-W. For Dataset 2, the MAE and MASE of MHA-TCN-W decrease by approximately 1.44%, respectively, compared with MHA-TCN, and by 12.55% respectively, compared with TCN-W.

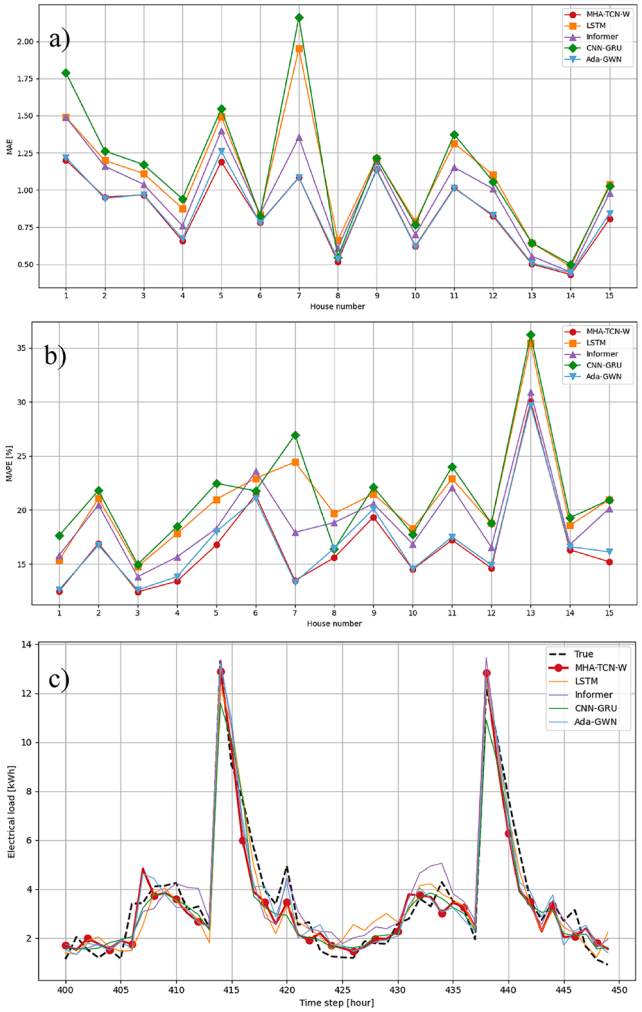


Fig. 7. Performance of models on dataset 1. a) MAE of 15 randomly selected buildings. b) MAPE of 15 randomly selected buildings. c) Visualization of the prediction results for the first building.

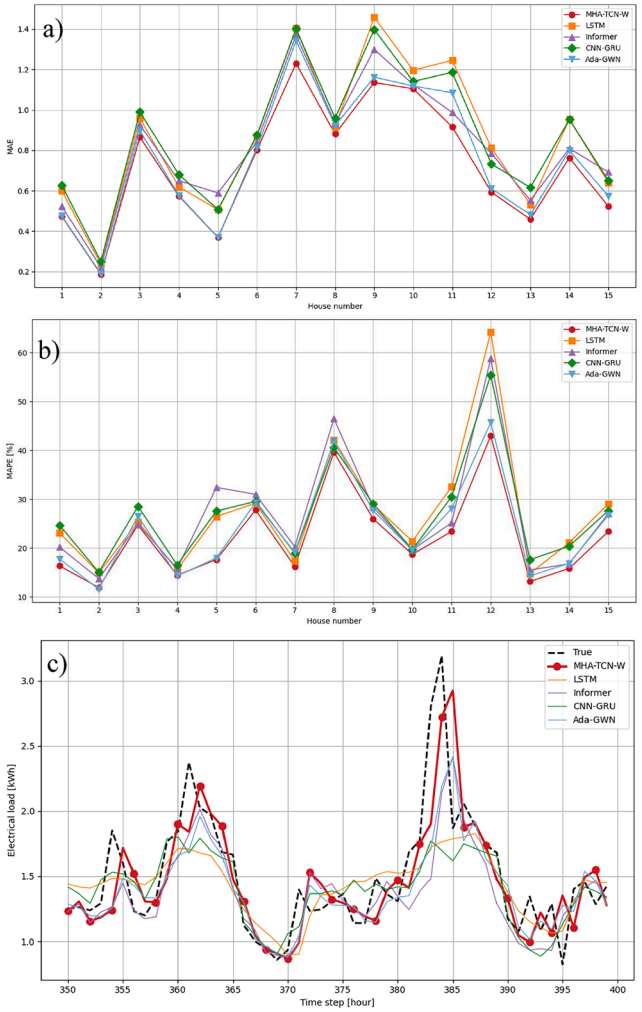


Fig. 8. Performance of models on dataset 2. a) MAE of 15 randomly selected buildings. b) MAPE of 15 randomly selected buildings. c) Visualization of the prediction results for the first building.

Figs. 5 and 6 provides a visual representation of the experimental results summarized in Table 3. Figs. 5(a) and 6(a) shows the MAE comparison results for 15 randomly selected residential buildings from dataset 1and 2, respectively. It shows that the MAE values of the MHA-TCN-W model are lower than those of the MHA-TCN and TCN-W models across those residential buildings. Figs. 5(b) and 6(b) presents the MAPE comparison for the same residential samples. The MAPE values of the MHA-TCN-W model are lower than those of the MHA-TCN model and TCN-W model. Additionally, the MAE and MAPE metrics exhibit an inverse trend across a subset of residential samples. This phenomenon arises from the distinct measurement characteristics of the two indicators. For residential units with higher load levels, even if the prediction error is large in absolute value, the relative error is comparatively smaller due to the higher actual load value in the denominator. Conversely, for residential units with lower load levels, although the MAE is smaller, the MAPE is amplified because of the lower actual load value, resulting in an opposite trend.

Figs. 5(c) and 6(c) displays the load prediction curve comparison for the first residential building from dataset 1 and 2, respectively. The prediction curve of the MHA-TCN-W model shows closer alignment with the actual load curve compared to MHA-TCN and TCN-W, capturing load variation trends with greater accuracy. Overall, the incorporation of the MHA module results in an average performance improvement of 14.01%, while the inclusion of the WaveNet module yields an average improvement of 3.63%. These findings indicate that each component in the MHA-TCN-W architecture contributes to overall performance improvement. In particular, the MHA mechanism effectively extracts and integrates homogeneous features, enabling the model to more accurately capture residential electricity consumption patterns and thereby contributing more to the overall performance gains.

4.5. Simulation results for scenario 3

Table 4 presents the simulation results for Scenario 3. The results indicate that the proposed method outperforms all baseline models across the evaluated metrics and demonstrating higher prediction accuracy on both datasets. Specifically, for Dataset 1, the model achieves reductions of 12.91% and 12.89% in MAE and MASE, respectively, compared with the second-best baseline model, Informer. For Dataset 2, compared with Informer, the MAE and MASE are reduced by 6.56% and 6.55%, respectively.Compared with the Ada-GWN algorithm, the proposed method attains comparable accuracy while exhibiting higher computational efficiency during both training and testing phases, as shown in Table 5. It should be noted that the proposed MHA-TCN-W model achieves higher prediction accuracy at the cost of computational efficiency. Therefore, its computational time is not directly comparable to that of simple models such as LSTM, Informer, and CNN-GRU. To ensure fairness, Table 5

Table 4
Calculation results of evaluation indicators for prediction.

Dataset	Method	Metrics			
		MAE	MAPE(%)	RMSE	MASE
1	LSTM	0.9900	23.78%	1.4599	0.4228
	Informer	0.8862	22.01%	1.2813	0.3793
	CNN-GRU	0.9823	23.62%	1.4546	0.4205
	Ada-GWN	0.7751	19.05%	1.1426	0.3318
	MHA-TCN-W	0.7718	18.90%	1.1416	0.3304
2	LSTM	0.9183	28.35%	1.3653	0.4448
	Informer	0.8947	27.41%	1.3233	0.4333
	CNN-GRU	0.9330	28.07%	1.3910	0.4519
	Ada-GWN	0.8509	26.09%	1.2637	0.4122
	MHA-TCN-W	0.8360	24.96%	1.2578	0.4049

Table 5
The computational time of each method.

Dataset	Method	Time [s]	
		Training Time	Testing Time
1	Ada-GWN	1955.94	0.61
	MHA-TCN-W	1605.48	0.59
2	Ada-GWN	395.98	0.16
	MHA-TCN-W	341.47	0.15

represents only the computational time of MHA-TCN-W and Ada-GWN, which share a comparable level of model complexity.

Figs. 7 and 8 visually presents the information from Table 4. Figs. 7(a) and 8(a) shows the MAE comparison results for 15 randomly selected residential buildings from dataset 1 and 2, respectively. The results indicate that MHA-TCN-W generally outperforms traditional deep learning models such as LSTM, Informer, and CNN-GRU, while achieving accuracy comparable to Ada-GWN. Figs. 7(b) and 8(b) presents the MAPE comparison results for the same residential samples. It can be observed that MHA-TCN-W maintains lower MAPE values, demonstrating stable performance even for residences with pronounced load fluctuations. Furthermore, Figs. 7(c) and 8(c) displays the load forecasting curve comparison for the first residential building from dataset 1 and 2, respectively. The MHA-TCN-W model exhibits closer alignment with the actual load curve in overall trend fitting and shows higher accuracy in peak load and fluctuation regions. In summary, MHA-TCN-W demonstrates strong prediction accuracy and stability, along with superior adaptability in handling peak and fluctuation characteristics. These results further confirm the method's effectiveness and robustness in temporal dependency modeling and homogeneous feature representation.

To evaluate the model stability, we conducted repeated experiments ($N = 3$) under different random seeds. The results show that while models exhibit slight variances (for MAE, MHA-TCN-W and Ada-GWN have standard deviations of ± 0.0014 and ± 0.0019 on Dataset 1, and both exhibit ± 0.0029 on Dataset 2), the absolute performance gaps between our model and the strongest baseline ($\Delta = 0.0033$ on Dataset 1, $\Delta = 0.0160$ on Dataset 2) are larger than the potential fluctuation margins. This confirms that the performance improvement of our proposed method is robust and statistically significant.

4.6. Summary of simulation analysis

The following conclusions can be drawn from the simulation results above:

1. The MHA-TCN-W model exhibits low sensitivity to hyperparameters, indicating it has good stability and robustness.

2. Each module in the MHA-TCN-W architecture contributes to the enhancement of predictive performance. In particular, the MHA mechanism in MHA-TCN-W can extract and integrate homogeneous features influencing user energy consumption patterns, enabling the model to more accurately characterize residential electricity usage patterns and thereby improve prediction accuracy.

3. MHA-TCN-W exhibits excellent performance in both prediction accuracy and stability, demonstrating strong adaptability in handling peak and load fluctuation characteristics. Compared with traditional neural network models such as LSTM, Informer, and CNN-GRU, MHA-TCN-W achieves notable improvements in prediction accuracy. Furthermore, when compared with the more complex Ada-GWN model, it maintains a high level of predictive accuracy while achieving superior computational efficiency.

5. Conclusion and future work

In this paper, we propose an MHA-TCN-W load forecasting framework that effectively extracts and integrates homogeneous features influencing user energy consumption patterns, enabling personalized batch forecasting. The MHA module within this framework efficiently extracts homogeneous features from multiple residential load and integrates them into the consumption data, allowing the model to more accurately characterize residential electricity usage patterns. Simultaneously, the MHA module preserves individual characteristics during feature fusion, enabling personalized batch forecasting. The enriched data undergoes TCN processing to extract temporal dependencies, thereby improving model prediction accuracy. Compared to baseline algorithms such as LSTM, Informer, CNN, and Ada-GWN, the proposed algorithm demonstrates superior performance in both load prediction accuracy and operational efficiency.

Future research will explore whether transfer learning can mitigate the issues of lengthy training times and substantial data requirements. In addition, we will incorporate advanced architectures such as GNN and global time-series models into the research framework and further optimize the forecasting model to enhance predictive performance.

CRediT authorship contribution statement

Jing Zhang: Writing – original draft, Methodology, Investigation, Conceptualization; **Xiaohui Li:** Writing – review & editing, Investigation; **Yuemin Ding:** Visualization, Supervision, Funding acquisition; **Bin Cai:** Writing – review & editing; **Jie He:** Investigation; **Jun Liu:** Supervision, Investigation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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